

Reinforcement Learning

Dynamic Programming Sutton/Barto* Chapter 4

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With figures from Sutton/Barto*

*Sutton and Barto, Reinforcement Learning: An Introduction,
2nd edition, MIT Press, Cambridge, MA, 2018



Online Material



Topics of this Course

- Introduction to reinforcement learning
- Markov decision processes
- **Part I: Tabular Methods**
 - **Dynamic programming**
 - Monte Carlo methods
 - Temporal-difference learning
 - Multi-step bootstrapping
 - Planning and learning with tabular methods
- **Part II: Approximate Solution Methods**
 - Prediction and Control using Approximation
 - Eligibility Traces
 - Policy Gradient Methods
- **Part III: Modern RL Methods**
 - Deep Reinforcement Learning
 - Current Applications

Summary of Notation

General

X	capital letters: random variables
x, p	lower-case letters: realizations of random variables or scalar functions
$\mathbf{w}$	Bold lower-case letters: real-valued vectors (even if random variables)
$\mathbf{W}$	bold capitals: matrices
α	Greek letters: parameters (vectors if in bold)
$\Pr\{X = x\}$	probability that a random variable X takes on the value x
$X \sim p$	random variable X selected from distribution $p(x) = \Pr\{X = x\}$
$\mathbb{E}[X]$	expectation of a random variable X , i.e., $\mathbb{E}[X] = \sum_x p(x)x$
$\operatorname{argmax}_a f(a)$	a value of action a at which $f(a)$ takes its maximal value

Value Function

G_t	return (cumulative reward) following time t
$G_{t:h}$	return from t to h (discounted and corrected)
$v_\pi(s)$	value of state s under policy π (expected return)
$v_*(s)$	value of state s under the optimal policy
$q_\pi(s, a)$	value of taking action a in state s under policy π
$q_*(s, a)$	value of taking action a in state s under the optimal policy
V, V_t	array estimates of state-value function v_π or v_*
Q, Q_t	array estimates of action-value function q_π or q_*

MDP

s, s'	states
a	an action
r	a reward
$\mathcal{S}$	set of all (nonterminal) states, $\mathcal{S}^+$ are all states
$\mathcal{A}(s)$	set of all actions available in state s
γ	discount-rate parameter
t	discrete time step
T	final time step of an episode (a.k.a. horizon)
A_t	random variable for the action at time t
S_t	random variable for the state at time t
R_t	random variable for the reward at time t
$p(s', r s, a)$	probability of transition to state s' and receiving reward r , from state s taking action a .
$p(s' s, a)$	probability of transition to state s' from state s taking action a .
$r(s, a)$	expected immediate reward from state s after action a .
$r(s, a, s')$	expected immediate reward from state s to s' with action a .
$\pi(a s)$	probability of taking action a in state s under stochastic policy π
$\pi(s)$	action taken in state s under deterministic policy π

Goal

- **Remember:** Finite MDPs have an optimal deterministic policy (under some mild conditions).
- **Goal:** Compute the **optimal deterministic policy** $\pi_* = \operatorname{argmax}_a q_*(s, a)$ given a perfect MDP model.
- Solve the Bellman optimality equations (the optimal policy always picks the best action):

$$v_*(s) = \max_a \mathbb{E}_\pi [R_{t+1} + \gamma v_*(S_{t+1}) | S_t = s, A_t = a]$$

or

$$q_*(s, a) = \max_a \mathbb{E}_\pi \left[R_{t+1} + \gamma \underbrace{\max_{a'} q_*(S_{t+1}, a')}_{= v_*(S_{t+1})} \middle| S_t = s, A_t = a \right]$$

Control Problem:
Find the optimal policy

- **Issues:** The value estimate for S_t depends on the current estimates of the successor states S_{t+1} ! We need to solve all optimality equations at the same time.

- **Idea:** We can find the optimal policy by stepwise **improving a policy** till it is optimal. A policy is better if it has larger state values.

$$\pi' \geq \pi \Leftrightarrow v_{\pi'}(s) \geq v_\pi(s) \quad \text{for all } s \in \mathcal{S}$$

- To do this we need to be able to compute v_π and $v_{\pi'}$.

Prediction Problem:
Estimate the value function for a policy.

Tabular Methods

- Tabular means: The value functions v or q are stored in arrays (tables) called V or Q .
- Important property: Entries in the table do not influence each other. We can change them independently.
- Tabular methods find the exact solution (i.e., optimal value function and policy) given enough memory and time.
- **Issues:**
 - Table size:
 - v and deterministic π have $|\mathcal{S}|$ entries
 - q has $|\mathcal{S}| \times |\mathcal{A}|$ entries

This is only feasible for problems with a small discrete state/action space!
 - No generalization: Knowledge about the value of one state (state-action pair) does not help us with similar states
- We will later talk about approximate methods that generalize and can work with large state spaces.

Value Function

s	State Value $V(s)$
(1,1)	0.7453
(1,2)	0.8016
...	...

Q-Value Function

s	a	$q(s, a)$
(1,1)	up	0.7453
(1,1)	right	0.6709
(1,1)	down	0.7003
(1,1)	left	0.7109
...	...	...

Policy

s	Action $\pi(s)$
(1,1)	up
...	...
...	...

Prediction vs. Control

- In this class, we will always talk about two tasks:
 1. **Prediction to evaluate a policy:** Estimate the value function for a given policy. We can use this to compare two policies.
 2. **Control:** Control the agent's behavior to find the optimal policy. This requires being able to evaluate policies.

Prediction: Policy Evaluation

What is the value function for a given policy?

Policy Evaluation to Estimate v_π

Goal: Compute the state-value function v_π for a given policy π .

Bellman expectation equations give us the recursive relationship:

$$\begin{aligned} v_\pi(s) &\stackrel{\text{def}}{=} \mathbb{E}_\pi[G_t | S_t = s] \\ &= \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} | S_t = s] \\ &= \mathbb{E}_\pi[R_{t+1} + \gamma v_\pi(S_{t+1}) | S_t = s] \\ &= \sum_a \pi(a|s) \sum_{s', r} p(s', r | s, a) [r + \gamma v_\pi(s')] \end{aligned}$$

with $|\mathcal{S}|$ equations, one for each state s .

Guarantees:

- Discounted finite MDPs ($\gamma < 1$) have a unique v_π .
- Undiscounted finite MDPs ($\gamma = 1$) have a unique v_π if π guarantees to reach a terminal state.
- Finite-horizon MDPs have unique v_π^k where k is the time step. I.e., value depends on how close we are to the end of the episode.

Issue: Recursive relationship between $v_\pi(s)$ and $v_\pi(s')$

Solution Option 1: Linear Program

Solve the system of $|\mathcal{S}|$ linear Bellman Expectation Equations:

$$v_{\pi}(s_1) = \sum_a \pi(a|s_1) \sum_{s',r} p(s',r | s_1, a) [r + \gamma v_{\pi}(s')]$$
$$v_{\pi}(s_2) = \sum_a \pi(a|s_2) \sum_{s',r} p(s',r | s_2, a) [r + \gamma v_{\pi}(s')]$$

... for all $s \in \mathcal{S}$

Formulate as a linear program

- Objective function: $\min_V \sum_{s \in \mathcal{S}} V(s)$
- Constraints: $V(s) \geq \sum_{s',r} p(s',r | s, \pi(s)) [r + \gamma V(s')] \quad \forall s \in \mathcal{S}$

Why this works: The smallest feasible solution will satisfy all Bellman equations.

$$V(s) = \sum_{s',r} p(s',r | s, \pi(s)) [r + \gamma V(s')] \quad \forall s \in \mathcal{S}$$

Example: 2-State MDP

Min: $V(s_1) + V(s_2)$

S.t.: $V(s_1) \geq R_1 + \gamma(P_{11}V(s_1) + P_{12}V(s_2))$

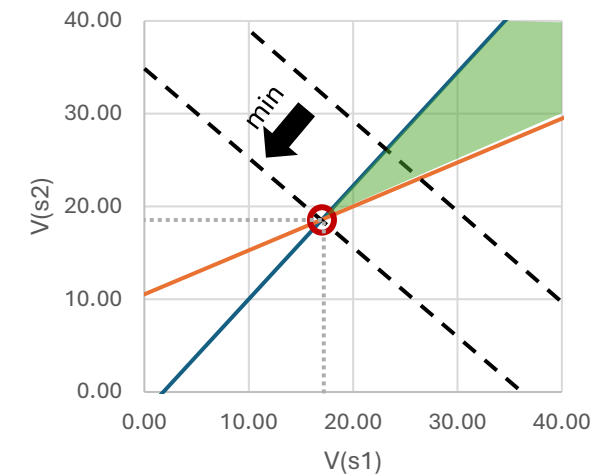
$V(s_2) \geq R_2 + \gamma(P_{21}V(s_1) + P_{22}V(s_2))$

Where **following** π has

R_x immediate reward of s_x

P_{xy} transition probability $s_x \rightarrow s_y$

LP for 2-State MDP



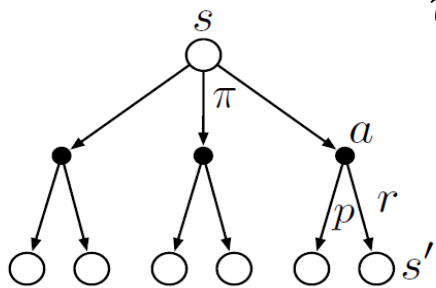
— Constraint $V(s_1)$ — Constraint $V(s_2)$
- - Objective Feasible Region

Solution Option 2: Iterative Policy Evaluation

Dynamic Programming: solves a complex optimization problem by breaking it down into smaller, overlapping subproblems and using their solutions to build the overall optimal solution.

- Start with arbitrary values for all $v(s)$
- Update all state values (called sweeping over the state space) using the **Bellman Update** (k is the iteration counter):

Backup diagram

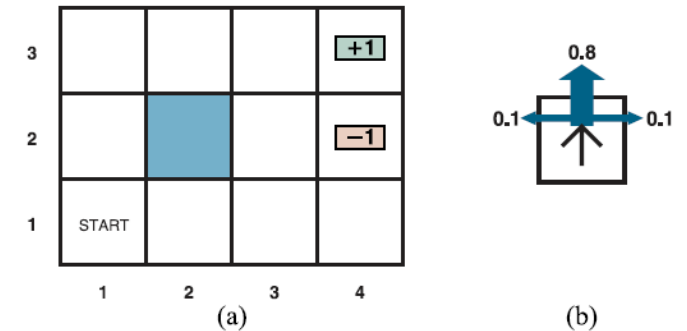


$$v_{k+1}(s) \stackrel{\text{def}}{=} [R_{t+1} + \gamma v_k(S_{t+1}) | S_t = s]$$
$$= \underbrace{\sum_a \pi(a|s) \sum_{s',r} p(s',r | s, a) [r + \gamma v_k(s')]}_{\text{In DP called "expected update."}}$$

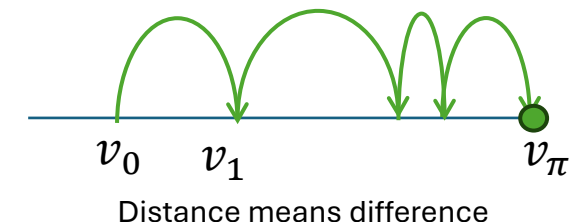
In DP called “expected update.”
Uses the expectation over all possible actions and next states.

- **Convergence:** Updates converge to v_π for $k \rightarrow \infty$ iterations. Reason is on the next slide.

Sweep the state space



Bellman updates



Convergence of Iterative Policy Evaluation

- The Bellman update can be rewritten as an operator:
Bellman Expectation Operator T^π :

$$(T^\pi v)(s) = \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a)[r + \gamma v(s')]$$

Notation:

$(T^\pi v)$ is the new value function after applying the operator. We could write $v_{k+1} = T^\pi(v_k)$

- The Fixed Point Property:** The value function is unchanged if all state values are already consistent.
 $v_\pi = T^\pi(v_\pi)$

- The Contraction Property:** The operator T^π is γ -contracting (see textbook for derivation):

$$\|T^\pi u - T^\pi v\|_\infty \leq \gamma \|u - v\|_\infty \text{ for any } \gamma < 1$$

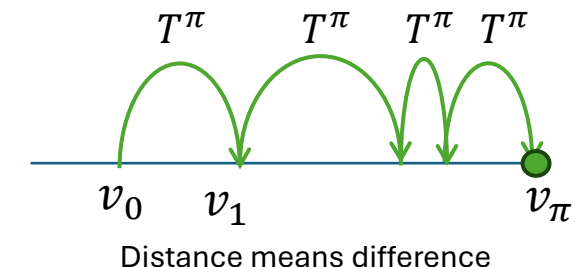
I.e. : Applying T^π reduces the largest difference (infinity norm) between any arbitrary value functions u and v by at least a factor of γ

- This implies:

$$\|v_\pi - T^\pi v\|_\infty \leq \gamma \|v_\pi - v\|_\infty \text{ for any } \gamma < 1$$

Convergence if $\gamma < 1$:
 $T^\pi T^\pi T^\pi \dots T^\pi v \rightarrow v_\pi$ for any v

Bellman updates



Alg. Iterative Policy Evaluation

Notation: V is the tabular estimate of v

Iterative Policy Evaluation, for estimating $V \approx v_\pi$

Input π , the policy to be evaluated

Algorithm parameter: a small threshold $\theta > 0$ determining accuracy of estimation

Initialize $V(s)$ arbitrarily, for $s \in \mathcal{S}$, and $V(\text{terminal})$ to 0

Loop:

$\Delta \leftarrow 0$

Loop for each $s \in \mathcal{S}$:

$v \leftarrow V(s)$

$V(s) \leftarrow \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a) [r + \gamma V(s')]$

$\Delta \leftarrow \max(\Delta, |v - V(s)|)$

until $\Delta < \theta$

Bellman operator T^π :
 $V_{k+1} = T^\pi V_k$

Stop when value function changes little:

$$\|T^\pi V_k - V_k\|_\infty < \theta$$

It can be shown that the error is bounded:

$$\|V_k - v_\pi\|_\infty < \frac{\theta}{1 - \gamma}$$

If we stop policy evaluation early (after n iterations) then it is called **modified policy iteration** and we end up with an approximation.

Control: Policy Iteration

Find the optimal policy based on the environment model.

Policy Improvement and the Greedy Policy

Policy Improvement Theorem

$$q_{\pi}(s, \pi'(s)) \geq v_{\pi}(s) \quad \forall s \in \mathcal{S} \Rightarrow v_{\pi'}(s) \geq v_{\pi}(s) \quad \forall s \in \mathcal{S}$$

If π' is the same as π but has a better action for at least one state, then π' is strictly better than π

- **Greedy policy improvement:** create a new policy π' by choosing in each state the best (aka greedy) action with respect to $q_{\pi}(s, a)$.

$$\pi'(s) = \operatorname{argmax}_a q_{\pi}(s, a)$$

- **Guarantee:** This **greedy policy** π' is strictly better unless π is already optimal.
- **Proof:**

$$\begin{aligned} v_{\pi_{k+1}}(s) &= q_{\pi_k}(s, \pi_{k+1}(s)) = q_{\pi_k}(s, \operatorname{argmax}_a q_{\pi_k}(s, a)) \\ &= \max_a q_{\pi_k}(s, a) \\ &\geq q_{\pi_k}(s, \pi_k(s)) \\ &\geq v_{\pi_k}(s) \end{aligned}$$

π_{k+1} chooses greedy action for s

Choosing the best action will never lead to a lower value!

Policy Iteration

Idea: Create a sequence of monotonically improving policies:

$$\pi_0 \xrightarrow{E} v_{\pi_0} \xrightarrow{I} \pi_1 \xrightarrow{E} v_{\pi_1} \xrightarrow{I} \pi_2 \xrightarrow{E} v_{\pi_2} \xrightarrow{I} \pi_3 \xrightarrow{E} v_{\pi_3} \xrightarrow{I} \dots \pi_* \xrightarrow{E} v_*$$

$\xrightarrow{E}$ policy eval. step
 $\xrightarrow{I}$ policy improvement step

Guarantees:

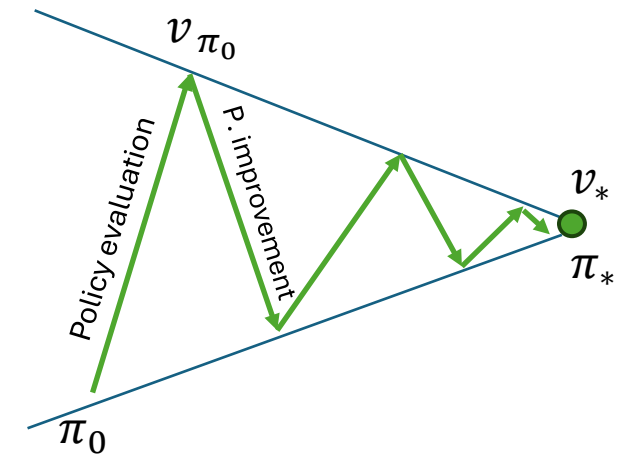
- Policy improvement is guaranteed to improve the policy.
- Finite MDPs have a finite number of policies $\rightarrow$ Policy Iteration must **converge**!

Advantages:

- Policy becomes stable before v_* is reached.
- Policy iteration typically converges fast.

Issue:

- Policy estimation is computationally expensive!
Using fewer iterations leads to modified policy iteration.



Alg. Policy Iteration

Policy Iteration (using iterative policy evaluation) for estimating $\pi \approx \pi_*$

1. Initialization

$V(s) \in \mathbb{R}$ and $\pi(s) \in \mathcal{A}(s)$ arbitrarily for all $s \in \mathcal{S}$; $V(\text{terminal}) \doteq 0$

2. Policy Evaluation

Loop:

$\Delta \leftarrow 0$

Loop for each $s \in \mathcal{S}$:

$v \leftarrow V(s)$

$V(s) \leftarrow \sum_{s',r} p(s',r|s,\pi(s)) [r + \gamma V(s')]$

$\Delta \leftarrow \max(\Delta, |v - V(s)|)$

until $\Delta < \theta$ (a small positive number determining the accuracy of estimation)

Just policy evaluation from before.
We can also stop this loop early
(modified policy iteration).

3. Policy Improvement

policy-stable $\leftarrow$ *true*

For each $s \in \mathcal{S}$:

old-action $\leftarrow \pi(s)$

$\pi(s) \leftarrow \operatorname{argmax}_a \sum_{s',r} p(s',r|s,a) [r + \gamma V(s')]$

If *old-action* $\neq \pi(s)$, then *policy-stable* $\leftarrow$ *false*

If *policy-stable*, then stop and return $V \approx v_*$ and $\pi \approx \pi_*$; else go to 2

Picks the greedy action with the highest
Q-value to improve the policy.

Control: Value Iteration

Find the optimal value function based on the environment model.

Value Iteration

- Combines policy improvement and truncated policy evaluation into a single step. This is called the **expected Bellman optimality update**.

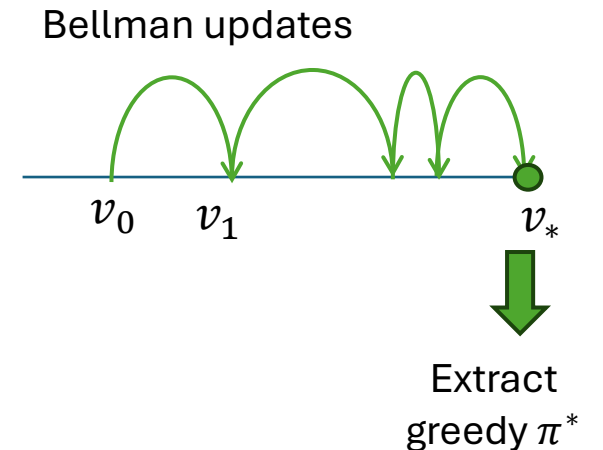
$$\begin{aligned} v_{k+1}(s) &\stackrel{\text{def}}{=} \max_a \mathbb{E}[R_{t+1} + \gamma v_k(S_{t+1}) | S_t = s, A_t = a] \\ &= \max_a \sum_{s', r} p(s', r | s, a) [r + \gamma v_k(s')] \end{aligned}$$

$\max_a$... policy improvement
 $\mathbb{E}$... truncated policy evaluation (1 step)

- Written as the **Bellman optimality operator**:

$$(T^*v)(s) = \max_a \sum_{s', r} p(s', r | s, a) [r + \gamma v(s')]$$

- Guarantee:** T^* is a non-linear operator, but it is also γ -contracting. The Sequence $\{v_k\}$ converges to v_* .



Alg. Value Iteration

Policy Iteration (using iterative policy evaluation) for estimating $\pi \approx \pi_*$

1. Initialization

$V(s) \in \mathbb{R}$ and $\pi(s) \in \mathcal{A}(s)$ arbitrarily for all $s \in \mathcal{S}$; $V(\text{terminal}) \doteq 0$

2. Policy Evaluation

Loop:

$\Delta \leftarrow 0$

Loop for each $s \in \mathcal{S}$:

$v \leftarrow V(s)$

$V(s) \leftarrow \sum_{s',r} p(s',r|s,\pi(s)) [r + \gamma V(s')]$

$\Delta \leftarrow \max(\Delta, |v - V(s)|)$

until $\Delta < \theta$

3. Policy Improvement

For each $s \in \mathcal{S}$:

For each $a \in \mathcal{A}(s)$:

$v_a \leftarrow \sum_{s',r} p(s',r|s,a) [r + \gamma V(s')]$

$\pi(s) \leftarrow \arg\max_a v_a$

If $\pi(s) \neq \text{old-action}$, then $\text{policy-stable} \leftarrow \text{false}$

If policy-stable , then stop and return $V \approx v_*$

Combine improvement and modified evaluation (1 iteration only) into a single update.

Value Iteration, for estimating $\pi \approx \pi_*$

Algorithm parameter: a small threshold $\theta > 0$ determining accuracy of estimation

Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $V(\text{terminal}) = 0$

Loop:

$\Delta \leftarrow 0$

Loop for each $s \in \mathcal{S}$:

$v \leftarrow V(s)$

$V(s) \leftarrow \max_a \sum_{s',r} p(s',r|s,a) [r + \gamma V(s')]$

$\Delta \leftarrow \max(\Delta, |v - V(s)|)$

until $\Delta < \theta$

Output a deterministic policy, $\pi \approx \pi_*$, such that

$\pi(s) = \arg\max_a \sum_{s',r} p(s',r|s,a) [r + \gamma V(s')]$

Bellman optimality operator:

$$V_{k+1} = T^* V_k$$

Stopping criterion from policy evaluation

Extract the greedy policy.

Code...

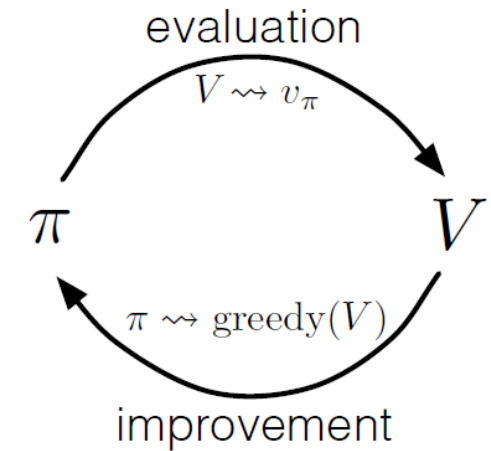
GPI: Generalized Policy Iteration

Generalizing value and policy iteration into a single framework.

GPI: Generalized Policy Iteration

Steps

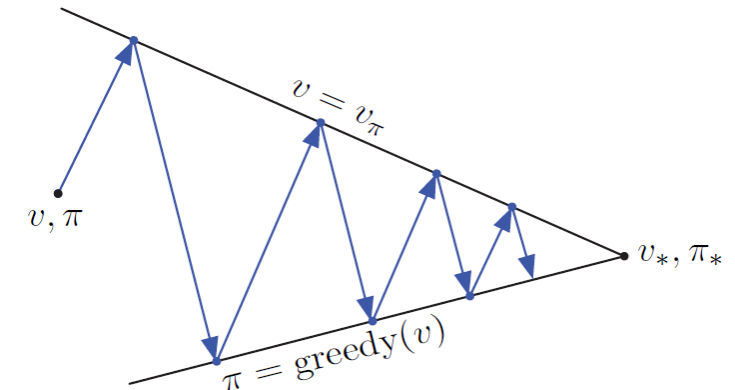
- **Policy evaluation:** make the value function consistent with the current policy π .
- **Policy improvement:** make policy greedy with respect to the current value function estimate V .



Most DP methods can be described as a special case of GPI:

- **Policy iteration:** each evaluation and improvement step is completed (by a sweep over $\mathcal{S}$) before moving on.
- **Value iteration:** uses a single iteration for the policy evaluation step.
- **Asynchronous DP:** Instead of sweeps over $\mathcal{S}$, updates states in some other order (e.g., focus on states where the value function changed a lot in the last step). This interleaves pol. eval. and pol. Improvement.

All three approaches typically converge to the optimal solution due to the **Policy Improvement Theorem**.



Issues with DP Methods

Require perfect knowledge of the model!

Space Complexity

- Tabular method: stores the complete value function as tables V or Q .

Time Complexity

- Updates repeatedly sweep over the whole state space.
- Wastes computation by updating states that are not relevant for the optimal behavior. I.e., the optimal policy will never visit them.
- Tabular methods can not generalize between similar states.

Asynchronous
DP can help

Efficiency

- **Direct Search:** In each state we can choose one of the available actions. Search space $O(|\mathcal{A}|^{|\mathcal{S}|})$ is **exponential** in the number of states $|\mathcal{S}|$.
- **LP (linear program):** Linearize the non-linear Bellman Optimality Equations by introducing constraints for all action/state pairs. Leads to a time complexity of $O(|\mathcal{S}|^3)$.
- **DP:** Time $O(|\mathcal{S}|^2 \times |\mathcal{A}|)$ is **quadratic** in $|\mathcal{S}|$.

Sparse transition matrices lead to better performance.

Conclusion: Search and LP are impractical. The number of actions is typically relatively small. **DP can be used today with millions of states.**

What You Need to Know



Requirements:

- **Model access:** DP methods need access to the **model's** function p .
- **Small discrete state/action space** since it is a tabular method.

Tasks:

- **Prediction:** Calculates the value function for a given policy called in DP policy evaluation.
- **Control:** Calculates the optimal policy for an MDP. Is not truly control, but an offline planning method.

Value Iteration:

- Convert the recursive Bellman equations into the Bellman update operators and sweep the state space till the optimal value function is found.
- **The Expected Bellman operator is γ -contracting and converges to the optimal value function** (requires either $\gamma < 1$ or a guarantee that the policy will reach a terminal state).

Policy Iteration:

- **Policy Improvement Theorem:** the policy strictly improves with each policy iteration step.
- Value iteration is a special case of modified policy iteration with a single policy evaluation iteration.
- All DP methods can be generalized using the GPI evaluation/improvement framework.